



Google's Duplex rolling out

Google has confirmed that its reservation making, conversational AI is rolling out to a select group of Pixel users. <https://dgit.in/dec18-43-1>



Amazon exposes customers

Amazon accidentally disclosed names and email addresses of its customers because of a reported technical error. <https://dgit.in/dec18-43-2>



It's more complex than it looks like

automation and complexity that supports this distributed design.

It performs these following rather complex tasks automatically:

- Partition documents across an arrangement of distinct shards (containers)
- In a multi-node cluster, distribute the documents to shards that resides across all of the nodes
- Balance shards across all nodes in a cluster to evenly manage the indexing and search load
- In case of replication, duplicate each shard to provide data redundancy and failover
- Route requests from any node in the cluster to specific nodes containing the specific data that you need
- Seamlessly add and integrate new nodes as you find the need to increase the size of your cluster
- Redistribute shards to automatically recover from the loss of a node

Now that we have a fair idea of what elasticsearch is and the concept it is based on, let us try to create a small (but powerful) search engine using it as the underlying technology.

SETTING UP ELASTICSEARCH

Prerequisites

Java - Elasticsearch is built on Java, and requires at least Java 8 to run. Oracle Java and the OpenJDK are only supported. The same JVM version needs to be used on all Elasticsearch nodes and clients.

It is recommended to install Java version 1.8.0_131 or a later version for optimum use. Please note that elasticsearch will refuse to start if a bad version of Java is used.

Hardware Platform

Elasticsearch can be run on own hardware or using the hosted Elasticsearch Service on Elastic Cloud, which is available on AWS and Google Cloud Platform.



Elastic Cloud is available on both

Installation

Let us now proceed with installing elasticsearch using .zip packages. The code to do so can be found here: <https://dgit.in/ES1>. These packages can be used to install Elasticsearch on any system and are the easiest package format to use when trying out Elasticsearch.

The latest stable version of Elasticsearch can be found on the Download Elasticsearch page.

Running Elasticsearch

To start elasticsearch from the command line, just use:

```
./bin/elasticsearch
```

By default, Elasticsearch runs in the foreground, prints its logs to the standard output (**stdout**), and can be stopped by pressing Ctrl-C. Log printing to **stdout** can be disabled using the **-q** or **--quiet** option on the command line.

You can test that the node is running by sending an HTTP get request to port 9200 on localhost.

```
http://localhost:9200/
```

Running as a daemon task

To run Elasticsearch as a daemon, specify **-d** on the command line, and record the process ID in a file using the **-p** option.

```
./bin/elasticsearch -d -p pid
```

Log are written in the `<home>/logs/` directory. To shut down the daemon, kill the process ID in the pid file:

```
kill `cat pid`
```

Elasticsearch Configuration

Elasticsearch loads its configuration from the `<home>/config/elasticsearch.`

yaml file. The format of this config is as follows, with three configuration files:

- `elasticsearch.yml` for configuring Elasticsearch
- `jvm.options` for configuring JVM settings
- `log4j2.properties` for configuring logging

While `elasticsearch.yml` has extensive settings configuration for fine tuning your application, it is beyond the scope of this tutorial to touch upon all of them. Suffice to say, the default .yml file contains enough inline comments to detail on each of these settings.

There are however a number of settings which need to be considered before going into production

Path settings

The data and logs directories are sub-folders of `<home>` directory. If these are left in their default locations, there is a high risk of them being deleted while upgrading Elasticsearch to a new version.

In production use, we will almost certainly want to change the locations of the data and log folder as shown here: <https://dgit.in/ES2>.

Cluster name

An elastic node can only join a cluster when it shares its `cluster.name` with all the other nodes in the cluster. The default cluster name is 'elasticsearch', but it should be changed to an appropriate name which describes the purpose of the cluster.

```
cluster.name: logging-prod
```

It is important to ensure different cluster names in different environments, otherwise you might end up with nodes joining the wrong cluster.

Node name

By default, Elasticsearch uses the a randomly generated UUID as the node id. It is worth configuring a more meaningful name which will also have the advantage of persisting after restarting the node.

Network host

Elasticsearch binds to loopback ad-